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$$\begin{aligned} & \frac{1}{6} c_Y^2 \overline{y}_d^{i2i3} y_u^{i1i4} + \frac{1}{16\pi^2} \left(\frac{1}{1296} \frac{1}{m_2^2} (27 c_Y^2 \overline{y}_d^{\text{pr}} y_d^{\text{ir}} \overline{y}_u^{i2i3} y_u^{pi4} (-1 + s_Y^2) - c_Y^2 \right. \\ & \quad (81 (4 s_Y^2 \overline{y}_u^{r13} y_u^{r14} (\overline{y}_d^{i2p} y_d^{i1p} + 3 \overline{y}_u^{i2p} y_u^{i1p}) + \overline{y}_u^{\text{pr}} \overline{y}_u^{i2i3} y_u^{pi4} y_u^{i1r} (1 + c_Y^2)) + \\ & \quad 3 y_u^{i1i4} (-9 \overline{y}_d^{i2p} y_d^{\text{rp}} \overline{y}_u^{r13} (-1 + s_Y^2) + 27 \overline{y}_u^{pi3} \overline{y}_u^{i2r} y_u^{\text{pr}} (1 + c_Y^2) + \\ & \quad 2 \overline{y}_u^{i2i3} (47 g_1^2 + 81 g_2^2 + 240 g_3^2 - 108 s_Y^2 \overline{y}_u^{\text{pr}} y_u^{\text{pr}})) + 14 g_1^2 \overline{y}_u^{pi3} y_u^{pi4} \delta_{i1i2}) + \\ & \quad 4 g_1^2 (5 s_Y^2 \overline{y}_d^{i2p} y_d^{i1p} - 13 c_Y^2 \overline{y}_u^{i2p} y_u^{i1p}) \delta_{i3i4} \Big) - \frac{1}{12} \sum c_Y g_1^2 \frac{1}{m_2} \overline{y}_u^{i2i3} \\ & \quad y_u^{i1i4} (c_2 c_Y c_Y - 2 s_2 c_Y s_Y) \text{LF}_{1,0}[\mathbf{m}_d^{\text{P}}] + \frac{243}{4} \sum g_1^4 \text{LF}_{3,0}[\mathbf{m}_d^{\text{P}}] \delta_{i1i2} \delta_{i3i4} - \\ & \quad \frac{5}{162} \sum g_1^4 \text{LF}_{4,-1}[\mathbf{m}_d^{\text{P}}] \delta_{i1i2} \delta_{i3i4} + \frac{16}{1215} \sum g_1^4 \text{LF}_{5,-2}[\mathbf{m}_d^{\text{P}}] \delta_{i1i2} \delta_{i3i4} - \\ & \quad \frac{1}{2} s_Y c_Y \frac{1}{m_2^4} \overline{y}_d^{\text{pr}} y_d^{\text{pr}} \overline{y}_u^{i2i3} y_u^{i1i4} (s_2 c_Y + s_Y c_Y) \text{LF}_{1,0}[\mathbf{m}_d^{\text{r}}] - \\ & \quad \frac{1}{12} \sum c_Y g_1^2 \frac{1}{m_2^4} \overline{y}_u^{i2i3} y_u^{i1i4} (c_2 c_Y c_Y - 2 s_2 c_Y s_Y) \text{LF}_{1,0}[\mathbf{m}_e^{\text{P}}] + \\ & \quad \frac{4}{81} \sum g_1^4 \text{LF}_{3,0}[\mathbf{m}_e^{\text{P}}] \delta_{i1i2} \delta_{i3i4} - \frac{5}{54} \sum g_1^4 \text{LF}_{4,-1}[\mathbf{m}_e^{\text{P}}] \delta_{i1i2} \delta_{i3i4} + \\ & \quad \frac{16}{405} \sum g_1^4 \text{LF}_{5,-2}[\mathbf{m}_e^{\text{P}}] \delta_{i1i2} \delta_{i3i4} - \\ & \quad \frac{1}{6} s_Y c_Y \frac{1}{m_2^4} \overline{y}_e^{\text{pr}} y_e^{\text{pr}} \overline{y}_u^{i2i3} y_u^{i1i4} (s_2 c_Y + s_Y c_Y) \text{LF}_{1,0}[\mathbf{m}_e^{\text{r}}] + \\ & \quad \frac{1}{12} c_Y \frac{1}{m_2^4} \overline{y}_u^{i2i3} y_u^{i1i4} (-2 s_Y \overline{y}_e^{\text{pr}} y_e^{\text{pr}} (s_2 c_Y + s_Y c_Y) + \sum g_1^2 (c_2 c_Y c_Y - 2 s_2 c_Y s_Y)) \text{LF}_{1,0}[\mathbf{m}_l^{\text{P}}] + \\ & \quad \frac{2}{81} \sum g_1^4 \text{LF}_{3,0}[\mathbf{m}_l^{\text{P}}] \delta_{i1i2} \delta_{i3i4} - \frac{5}{108} \sum g_1^4 \text{LF}_{4,-1}[\mathbf{m}_l^{\text{P}}] \delta_{i1i2} \delta_{i3i4} + \\ & \quad \frac{8}{405} \sum g_1^4 \text{LF}_{5,-2}[\mathbf{m}_l^{\text{P}}] \delta_{i1i2} \delta_{i3i4} - \frac{1}{12} c_Y \frac{1}{m_2^4} \overline{y}_u^{i2i3} y_u^{i1i4} \\ & \quad (6 s_Y \overline{y}_d^{\text{pr}} y_d^{\text{pr}} (s_2 c_Y + s_Y c_Y) + 6 \overline{y}_u^{\text{pr}} y_u^{\text{pr}} (c_Y^3 - s_Y^2 c_Y s_Y) + \sum g_1^2 (c_2 c_Y c_Y - 2 s_2 c_Y s_Y)) \\ & \quad \text{LF}_{1,0}[\mathbf{m}_q^{\text{P}}] + \frac{243}{4} \sum g_1^4 \text{LF}_{3,0}[\mathbf{m}_q^{\text{P}}] \delta_{i1i2} \delta_{i3i4} - \\ & \quad \frac{5}{324} \sum g_1^4 \text{LF}_{4,-1}[\mathbf{m}_q^{\text{P}}] \delta_{i1i2} \delta_{i3i4} + \frac{16}{1215} \sum g_1^4 \text{LF}_{5,-2}[\mathbf{m}_q^{\text{P}}] \delta_{i1i2} \delta_{i3i4} + \\ & \quad \frac{1}{6} \sum c_Y g_1^2 \frac{1}{m_2^4} \overline{y}_u^{i2i3} y_u^{i1i4} (c_2 c_Y c_Y - 2 s_2 c_Y s_Y) \text{LF}_{1,0}[\mathbf{m}_0^{\text{P}}] + \\ & \quad \frac{16}{243} \sum g_1^4 \text{LF}_{3,0}[\mathbf{m}_0^{\text{P}}] \delta_{i1i2} \delta_{i3i4} - \frac{10}{81} \sum g_1^4 \text{LF}_{4,-1}[\mathbf{m}_0^{\text{P}}] \delta_{i1i2} \delta_{i3i4} + \\ & \quad \frac{64}{1215} \sum g_1^4 \text{LF}_{5,-2}[\mathbf{m}_0^{\text{P}}] \delta_{i1i2} \delta_{i3i4} - \frac{1}{2} c_Y \frac{1}{m_2^4} \overline{y}_u^{\text{pr}} \overline{y}_u^{i2i3} y_u^{\text{pr}} y_u^{i1i4} (c_Y^3 - s_Y^2 c_Y s_Y) \text{LF}_{1,0}[\mathbf{m}_0^{\text{r}}] + \\ & \quad \frac{1}{24} c_Y \frac{1}{m_2^4} \overline{y}_u^{i2i3} y_u^{i1i4} (c_Y (g_1^2 (1 - 3 c_2 c_Y^2) - 3 g_2^2 (-1 + c_2 c_Y^2)) + 3 s_4 c_Y s_Y (g_1^2 + g_2^2)) \\ & \quad \text{LF}_{1,0}[\mathbf{m}_0] + \frac{1}{24} \frac{1}{m_2} c_Y^2 (3 \overline{y}_u^{i2i3} y_u^{pi4} (-s_Y^2 \overline{y}_d^{\text{pr}} y_d^{\text{ir}} + c_Y^2 \overline{y}_u^{\text{pr}} y_u^{\text{ir}}) - \\ & \quad y_u^{i1i4} (2 \overline{y}_u^{i2i3} (g_1^2 + 3 g_2^2) + 3 s_Y^2 \overline{y}_d^{i2p} y_d^{\text{rp}} \overline{y}_u^{r13} - 3 c_Y^2 \overline{y}_u^{pi3} \overline{y}_u^{i2r} y_u^{\text{pr}})) \text{LF}_{1,1}[\mathbf{m}_0] + \\ & \quad \frac{1}{216} (18 s_Y^2 c_Y^2 \overline{y}_d^{\text{pr}} y_d^{\text{ir}} \overline{y}_u^{i2i3} y_u^{pi4} + 2 s_Y^2 \overline{y}_d^{i2p} (9 c_Y^2 \overline{y}_u^{r13} (-6 y_d^{i1p} y_u^{r14} + y_d^{\text{rp}} y_u^{i1i4}) + \\ & \quad 8 g_1^2 y_d^{i1p} \delta_{i3i4}) + c_Y^2 (9 \overline{y}_u^{i2i3} y_u^{i1i4} (g_1^2 + 3 g_2^2) - \\ & \quad 4 g_1^2 \overline{y}_u^{pi3} y_u^{pi4} \delta_{i1i2} - 4 \overline{y}_u^{i2p} y_u^{i1p} (27 s_Y^2 \overline{y}_u^{r13} y_u^{r14} + 8 g_1^2 \delta_{i3i4})) \Big) \text{LF}_{1,2}[\mathbf{m}_0] + \\ & \quad \frac{1}{36} (s_Y^2 \overline{y}_d^{i2p} y_d^{i1p} (9 c_Y^2 \overline{y}_u^{r13} y_u^{r14} - 2 g_1^2 \delta_{i3i4}) - \\ & \quad c_Y^2 (g_1^2 \overline{y}_u^{pi3} y_u^{pi4} \delta_{i1i2} + \overline{y}_u^{i2p} y_u^{i1p} (9 c_Y^2 \overline{y}_u^{r13} y_u^{r14} - 2 g_1^2 \delta_{i3i4}))) \text{LF}_{2,1}[\mathbf{m}_0] + \\ & \quad \frac{1}{324} g_1^2 (9 c_Y^2 \overline{y}_u^{pi3} y_u^{pi4} \delta_{i1i2} + 2 (9 s_Y^2 \overline{y}_d^{i2p} y_d^{i1p} - 9 c_Y^2 \overline{y}_u^{i2p} y_u^{i1p} + 4 g_1^2 \delta_{i1i2}) \delta_{i3i4}) \\ & \quad \text{LF}_{3,0}[\mathbf{m}_0] - \frac{5}{108} g_1^4 \text{LF}_{4,-1}[\mathbf{m}_0] \delta_{i1i2} \delta_{i3i4} + \frac{405}{8} g_1^4 \text{LF}_{5,-2}[\mathbf{m}_0] \delta_{i1i2} \delta_{i3i4} + \\ & \quad \frac{81}{2} g_1^4 \text{LF}_{3,0}[\tilde{\mu}] \delta_{i1i2} \delta_{i3i4} + \frac{1}{27} g_1^4 \text{LF}_{4,-1}[\tilde{\mu}] \delta_{i1i2} \delta_{i3i4} - \\ & \quad \frac{16}{405} g_1^4 \text{LF}_{5,-2}[\tilde{\mu}] \delta_{i1i2} \delta_{i3i4} + \frac{1}{216} g_1^2 c_Y^2 \overline{y}_u^{i2i3} y_u^{i1i4} \text{LF}_{1,1,0}[\mathbf{m}_1, \mathbf{m}_q^{i1}] - \\ & \quad \frac{1}{432} g_1^2 \frac{1}{m_2} c_Y^2 \overline{y}_u^{i2i3} y_u^{i1i4} \text{LF}_{2,1,-1}[\mathbf{m}_1, \mathbf{m}_q^{i1}] + \frac{1}{1944} g_1^4 \text{LF}_{2,1,0}[\mathbf{m}_1, \mathbf{m}_q^{i1}] \delta_{i1i2} \delta_{i3i4} + \\ & \quad \frac{1}{1944} g_1^4 \text{LF}_{2,2,-1}[\mathbf{m}_1, \mathbf{m}_q^{i1}] \delta_{i1i2} \delta_{i3i4} - \frac{1}{972} g_1^4 \text{LF}_{3,1,-1}[\mathbf{m}_1, \mathbf{m}_q^{i1}] \delta_{i1i2} \delta_{i3i4} + \\ & \quad \frac{1}{1944} g_1^4 \text{LF}_{4,1,-2}[\mathbf{m}_1, \mathbf{m}_q^{i1}] \delta_{$$